# Deno$_n$MAE: Parameter-Efficient Recursive Denoising for Automatic Modulation Classification

Taha Bouhsine
*Independent Researcher*
contact@tahabouhsine.com

***Abstract*—Self-supervised pretraining with masked autoencoders (MAE) has recently advanced automatic modulation classification (AMC), notably in the DenoMAE family, by jointly denoising multiple signal modalities. We make three contributions. First, we replace the stack of distinct transformer layers with a single *recursive* block applied $n$ times, in the spirit of the Universal Transformer and ALBERT cross-layer parameter sharing, reducing the encoder parameter count by roughly a factor of the depth; we call the resulting model Deno$_n$MAE. Second, we add a *CLS-conditioned adaptive-depth gate*: each block reads the CLS token, emits a sigmoid scalar per sublayer, and scales its own residual update, so the recursive encoder can soft-skip later steps; a light ponder cost makes the gate learn how much depth the task actually needs. Third, we study *when* a multimodal RF representation should be learned by aligning modalities (contrastive) versus reconstructing them (denoising autoencoding), and show that a single forward pass can do both at once through per-modality CLS tokens. On a synthetic eight-class benchmark, the multi-task model (reconstruction plus a SigLIP contrastive head) denoises the signal to half the noise floor and gives the best AMC across SNR, while InfoNCE and SigLIP contrastive encoders fail to reconstruct the signal and exhibit a persistent modality gap. The gate, with the ponder cost, prunes the recursive encoder to roughly one effective layer at no accuracy cost. We frame the alignment-versus-reconstruction result through partial information decomposition: the denoising modalities are complementary, so the signal lives in their union, which contrastive alignment cannot represent.**

***Index Terms*—automatic modulation classification, masked autoencoders, universal transformers, ALBERT, parameter sharing, self-supervised learning, contrastive learning, modality gap, multimodal representation learning.**

## I. Introduction

Automatic modulation classification (AMC) identifies the modulation scheme of a received radio signal and is a building block of cognitive radio, spectrum monitoring, and electronic warfare [1]. Deep learning has become the dominant approach, and the field has moved from convolutional classifiers over raw I/Q toward self-supervised pretraining that exploits unlabeled signals. The DenoMAE family [8], [9] is a recent example: a multimodal masked autoencoder is pretrained to reconstruct clean signals and constellation diagrams from noisy inputs, then fine-tuned for AMC, reaching state-of-the-art accuracy with less labeled data. This paper adapts DenoMAE's multimodal denoising *motivation* rather than its exact recipe: we use two modalities (waveform and constellation) instead of five, a denoising objective with no input masking instead of 75% token masking, and a frozen linear probe over eight classes instead of a fine-tuned ten-class head. Within that simplified setting we ask what it takes to make multimodal denoising (i) parameter-efficient enough for edge deployment and (ii) explicit about *when* reconstruction, rather than contrastive alignment, is the objective to reach for in the first place.

Two questions remain underexplored. The first is *architectural efficiency*. AMC frequently runs on edge receivers with tight memory and power budgets, yet transformer encoders allocate a distinct parameter block per layer. The second is *objective choice*. Self-supervised RF pretraining can either *align* modalities in a shared embedding space, as in contrastive multimodal learning (CLIP [34], SigLIP [35]), or *reconstruct* them, as in DenoMAE. Practitioners often default to contrastive learning because it is modular and scales, but it is not obvious that alignment is the right objective for denoising-oriented modalities.

We address both. On architecture, we adopt a *recursive* transformer: one block whose weights are reused across depth, following the Universal Transformer [12] and ALBERT cross-layer sharing [13]. This preserves the effective depth of an $L$-layer model while cutting encoder parameters by about $L\times$. On objective, we show that the modalities used in denoising are *complementary* rather than redundant, and that contrastive alignment is structurally unable to recover the signal they jointly determine, regardless of whether the softmax (InfoNCE) or sigmoid (SigLIP) form is used.

Our contributions are:

- A recursive (weight-tied) transformer encoder for multimodal AMC that matches a depth-$L$ stack with roughly $1/L$ of the encoder parameters.
- A CLS-conditioned adaptive-depth gate that scales each block's residual update by a learned, input-dependent sigmoid scalar; with a ponder cost it learns how much recursive depth the task needs, pruning to about one effective layer here at no accuracy cost.
- A single-pass multi-task objective that combines denoising reconstruction with a SigLIP contrastive head over per-modality CLS tokens, which gives the best AMC of all variants.
- An analysis, grounded in partial information decomposition, of *when* contrastive alignment fails on RF modalities and *why* reconstruction succeeds, on a controlled benchmark that separates the objectives on identical data.

These are not co-equal in evidence. The objective contributions (third and fourth) are demonstrated on the benchmark below and are the core of this paper; the architectural contributions (first and second) are *proposals* whose validation — an untied baseline at equal width, a recursion-depth sweep, FLOP/latency counts, and a gate input-variance analysis — this synthetic benchmark cannot supply, and we scope them honestly as such (§VI).

## II. Related Work

### A. Deep learning for AMC

Classical AMC relied on hand-designed statistics: likelihood tests and cyclostationary / spectral-correlation features [3], [4] that are robust to channel effects but demand expert engineering. Deep learning replaced these with convolutional and recurrent classifiers over raw I/Q, established on the RadioML benchmarks [1], [2]; constellation-image classifiers and hybrid time/image models followed, and efficient FPGA inference of such classifiers is an active target for edge receivers [5]. Self-supervised pretraining, and masked autoencoding in particular, reduced the labeled-data requirement; DenoMAE [8] pretrains a multimodal masked autoencoder over five modalities (noisy and clean signal, noisy and clean constellation image, and noise as an explicit modality) with a 75% mask ratio, then fine-tunes for AMC; DenoMAE 2.0 [9] adds a position-aware local-patch classification objective and reports improved RadioML transfer over the original DenoMAE at matched SNR, with accuracy degrading gracefully toward chance at very low SNR. Both are ViT-Base image models (an $\approx 85$ M-parameter encoder over rendered constellation images), which sets the parameter-efficiency bar our recursive encoder targets. Self-supervised *contrastive* pretraining has also been applied to AMC, both standalone [6] and with transformers [7], which makes the align-versus-reconstruct question we study directly relevant to current practice; indeed Mod-CL [39] argues that task-agnostic contrastive pretexts are misaligned with AMC, which our objective analysis makes precise.

### B. Masked autoencoders

Masked autoencoders [10] learn representations by reconstructing masked inputs and transfer well to downstream tasks. MultiMAE [11] extends this to multiple modalities with a shared encoder, which is the inductive bias we build on: a single space that fuses modalities rather than two encoders aligned post hoc.

### C. Recursive and weight-tied transformers

The Universal Transformer [12] applies a single transformer function recurrently, tying weights across computation steps, and pairs it with Adaptive Computation Time (ACT [17]) to halt per token. ALBERT [13] shares parameters across layers in BERT, cutting model size with little accuracy loss. Looped transformers [20] (a theoretical programmable-computer construction) and relaxed recursive transformers [21] continue this line, the latter relaxing exact tying with layer-wise low-rank residuals. Closest to our setup, Mixture-of-Recursions [22] pairs a weight-tied recursive stack with per-token routers that assign recursion depth adaptively under a compute budget; our CLS-conditioned gate is a coarser, single-scalar version of the same recursion-plus- adaptive-depth idea. CoTFormer [23] interleaves recursive application with budget-adaptive computation, and the Sparse Universal Transformer [24] augments the weight-tied UT with mixture-of-experts and an ACT ponder penalty. All decouple depth from parameter count.

### D. Conditional computation and adaptive depth

A large literature already allocates depth or compute per input. ACT [17] introduced the *ponder cost* we borrow and learns a discrete halting decision; PonderNet [18] later recast halting probabilistically; Mixture-of-Depths [19] routes a token through or around a block; depth-adaptive transformers [25] exit early per token; and spatially adaptive computation time [26] halts per spatial region in residual networks. Our gate sits in this neighborhood but is deliberately simpler: a *smooth* sigmoid scalar, conditioned on a CLS summary, that scales a sublayer's residual update rather than making a discrete route/halt decision. Its pieces are individually standard. Input-conditioned sigmoidal gating of both sublayers with an identity-favoring initialization is GTrXL [27]; reading the gating signal from a pooled or CLS token rather than per token is the early-exit mechanism of DeeBERT [28] and PABEE [29]; static residual-scaling brackets it from the other side (Highway [14] gates, ReZero [15] and LayerScale [16] scale by learned but *input-independent* scalars); and the ponder cost is ACT [17]. Our scalar is input dependent like MoD/PonderNet but coarser (one scalar per sublayer, not per token) and smooth rather than discrete. We do not claim a new niche, only the combination of these known pieces inside a weight-tied recursion.

### E. Combining reconstruction and contrastive objectives

Joint reconstruction and contrastive training has been explored in audio-visual learning (CAV-MAE [30] and its finer-grained successor CAV-MAE Sync [33]) and multimodal MAE (M3AE [31]). Keeping the contrastive tokens modality-pure inside a single shared encoder is the idea behind Zorro [32], which uses attention masks to route each modality to its own representation while a fusion track attends to everything. We combine these for RF: both losses share one denoising forward pass, and a Zorro-style attention mask keeps each per-modality CLS token pure for the contrastive term while the signal tokens fuse both modalities for reconstruction.

### F. Contrastive learning and the modality gap

Contrastive objectives maximize agreement between paired views and underpin CLIP [34] and SigLIP [35]. They are known to leave a *modality gap*: image and text embeddings occupy separated cones in the shared space [36]. Partial information decomposition [40], [41] formalizes multimodal information as redundancy, uniqueness, and synergy; we use it to explain why contrastive alignment, which captures redundancy, is the wrong tool for complementary modalities.

## III. METHOD

### A. Problem setup

Adapting DenoMAE's denoising motivation, the receiver observes two complementary, lossy views of a clean length-$M$ symbol sequence $Z$ drawn from one of $K$ modulation schemes: a noisy *waveform* $X_{\text{wav}}$ that preserves symbol order, and a *constellation* histogram $X_{\text{con}}$ that preserves cluster geometry but discards order. Recovering $Z$ needs both, which is what makes the two modalities complementary rather than redundant; this complementarity drives the align-versus-reconstruct result throughout the paper. Appendix A gives the formal projections and the full benchmark construction.

### B. Recursive transformer encoder

We tokenize each modality. The waveform becomes $M$ tokens, one per symbol, $\boldsymbol{w}_i = \mathbf{E}_{\text{wav}}\,[\Re z_i, \Im z_i]^\top + \boldsymbol{p}_i^{\text{wav}}$; the constellation is split into $P \times P$ patches, each linearly embedded with a positional code. A learned `[CLS]` token is prepended.

The core is a single transformer block $f_\theta$ with one set of parameters $\theta$, applied $L$ times (the *recursion depth*; this $L$ is the subscript $n$ in the name Deno$_n$MAE):

$$\boldsymbol{h}^{(0)} = \text{tokens}, \qquad \boldsymbol{h}^{(\ell)} = f_\theta\Big(\boldsymbol{h}^{(\ell-1)}\Big), \quad \ell = 1, \dots, L, \tag{1}$$

where $f_\theta$ is pre-norm multi-head self-attention followed by a feed-forward network with residual connections. Unlike a standard stack, $\theta$ is *shared across all* $L$ *applications* (Universal-Transformer/ALBERT tying) and, within a model, across both modality towers. An $L$-deep encoder therefore costs the parameters of one block. For a block with $d$ model width, the encoder has $O(d^2)$ parameters independent of $L$, versus $O(Ld^2)$ for an untied stack.

### C. CLS-conditioned adaptive-depth gate

A weight-tied stack still runs $L$ applications. We let the model down-weight the later ones. At each recursion step the block reads the running CLS token $\boldsymbol{h}_0^{(\ell-1)}$ and emits two sigmoid scalars, one for the attention sublayer and one for the feed-forward sublayer:

$$\boldsymbol{g}^{(\ell)} = \sigma\big(\mathbf{W}_2\,\mathrm{GELU}(\mathbf{W}_1 \boldsymbol{h}_0^{(\ell-1)}) + \boldsymbol{b}\big) \in (0,1)^2. \tag{2}$$

Each sublayer's residual update is scaled by its gate, e.g. $\boldsymbol{h} \leftarrow \boldsymbol{h} + g_{\text{attn}}^{(\ell)}\,\mathrm{Attn}(\boldsymbol{h})$, so the block can soft-skip itself at step $\ell$. Two clarifications on what the scalar is and is not. It is a *global* scalar per sublayer, coarser than LayerScale (per channel) and ACT/MoD (per token); the conditioning token $\boldsymbol{h}_0^{(\ell-1)}$ in the multi-task model is $\texttt{CLS}_{\text{wav}}$, so the depth budget for the whole fused encoder is currently a function of the waveform summary alone (pooling both CLS tokens is the obvious symmetric fix and is left to the full-scale study). The gate MLP $(\mathbf{W}_1, \mathbf{W}_2)$ adds a small parameter count not charged against the tying saving (negligible at $d$=48, but real). The gate is initialized open ($\boldsymbol{b}$ set so $\sigma(\cdot) \approx 0.95$), an identity-favoring start in the spirit of GTrXL's gated-identity initialization [27]; training therefore begins slightly damped rather than at a clean identity. To turn the gate into a usable budget we add a ponder cost, $\lambda_{\text{p}}\,\frac{1}{2L}\sum_\ell \sum_{s\in\{\text{attn,ffn}\}} g_s^{(\ell)}$, which pushes gates toward zero unless the loss needs them open.

### D. Objectives

**Contrastive alignment.** Per-modality `[CLS]` embeddings $\boldsymbol{z}^{\text{wav}}, \boldsymbol{z}^{\text{con}}$ are aligned with an off-the-shelf contrastive loss: either the softmax InfoNCE of CLIP [34] or the pairwise-sigmoid loss of SigLIP [35]. We use both unmodified and restate their exact forms in Appendix A; the choice of softmax versus sigmoid does not change our conclusions.

**Joint reconstruction.** A per-token head reconstructs the clean signal from the fused, attended context, $\mathcal{L}_{\text{rec}} = \frac{1}{M}\sum_i \|g_\phi(\boldsymbol{h}_i^{(L)}) - z_i\|_2^2$. Because reconstruction is per token it never compresses $M$ symbols through a fixed-width latent.

**Single-pass multi-task.** Both objectives share *one* forward pass. We follow the DenoMAE family in calling this a (masked) autoencoder, but to be precise our model performs *denoising*, not masked, autoencoding: there is no input-token dropping and no mask ratio. The only mask in the model is the *attention* mask below, a Zorro-style modality-pure routing [32] that keeps the two CLS tokens modality-pure. The token sequence is $[\texttt{CLS}_{\text{wav}}, \texttt{CLS}_{\text{con}}$, wav tokens, con patches$]$ with an attention mask such that $\texttt{CLS}_{\text{wav}}$ attends only to itself and the waveform tokens, and $\texttt{CLS}_{\text{con}}$ only to itself and the constellation patches, while every signal token attends to all. The two CLS embeddings thus stay modality-pure for the contrastive term, and the signal tokens still fuse both modalities for reconstruction. The total loss is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_{\text{c}}\,\mathcal{L}_{\text{SigLIP}} + \lambda_{\text{p}}\,\mathcal{L}_{\text{ponder}}, \tag{3}$$

with $\lambda_{\text{c}}$=0.1, $\lambda_{\text{p}}$=0.03.

## IV. EXPERIMENTS

### A. Setup

We use the eight-scheme synthetic benchmark detailed in Appendix A ($K$=8 schemes, $M$=96 symbols, a $G$=24 constellation, mixed training SNR in $[-2, 10]$ dB). The recursion depth is $L$=4 with $d$=48 and 4 heads. The four models (InfoNCE, SigLIP, denoising reconstruction, and the multi-task combination) all share the same gated recursive encoder and differ only in the objective. The two contrastive models run the encoder as two separate passes, one per modality tower, and concatenate the resulting CLS embeddings; the reconstruction and multi-task models run the single attention-masked pass of Sec. III-D. Downstream readouts are *linear probes* (ridge) on frozen features; AMC features concatenate both towers for the contrastive models and use the fused encoder for the reconstruction and multi-task models.

### B. Signal recovery

Table I reports reconstruction MSE against the clean signal at 4 dB. The contrastive encoders sit at 0.50, far above the channel noise floor: their aligned embeddings discard the per-symbol detail a reconstruction needs. Both the reconstruction and multi-task models reach 0.10, half the noise floor, i.e. they

TABLE I

RECOVERING THE CLEAN SIGNAL (RECONSTRUCTION MSE) AND MODULATION CLASS (LINEAR PROBE AT 4 DB AND 12 DB). EIGHT CLASSES; CHANCE = 12.5%. ALL MODELS SHARE THE SAME GATED RECURSIVE-TRANSFORMER ENCODER AND DIFFER ONLY IN THE TRAINING OBJECTIVE.

| | recover-$Z$ | AMC acc. ↑ | |
|---|---|---|---|
| | MSE ↓ | 4 dB | 12 dB |
| noisy input (noise floor) | 0.199 | — | — |
| InfoNCE (contrastive) | 0.50 | 37% | 43% |
| SigLIP (contrastive) | 0.50 | 29% | 38% |
| denoising reconstruction | **0.10** | 41% | 58% |
| multi-task (recon + InfoNCE) | **0.10** | 39% | 52% |
| **multi-task (recon + SigLIP)** | **0.10** | **49%** | **73%** |

genuinely denoise. Fig. 6 shows the recovered symbols snapping from the noisy scatter back onto the clean constellation.

### C. Multi-task is best

The multi-task model, which adds the SigLIP CLS head to reconstruction in a single denoising pass, matches reconstruction MSE (0.10) while improving AMC over every other variant: 49% at 4 dB versus 41% for reconstruction-only and 37% for InfoNCE, and 73% versus 58% at 12 dB. The contrastive head, which is near-useless on its own here, becomes a helpful auxiliary once the encoder is also forced to reconstruct, organizing the shared class structure without costing reconstruction quality. The choice of head matters: swapping the SigLIP head for an InfoNCE head (Table I, "recon + InfoNCE") keeps the same reconstruction MSE but gives clearly worse AMC (39% vs 49% at 4 dB, 52% vs 73% at 12 dB). Sec. IV-F explains why: the InfoNCE head aligns the two modalities far more aggressively, and that aggressive alignment is exactly what hurts.

### D. AMC across SNR

Fig. 2 reports the DenoMAE-style evaluation: linear-probe modulation accuracy from $-4$ to 12 dB. The multi-task model leads across the entire range, the reconstruction model is second, and both contrastive encoders trail. All methods are far above chance, confirming that modulation class is *shared* information any representation can extract; the decisive separation is on reconstruction, where only the autoencoding objectives succeed.

### E. The gate prunes effective depth

Fig. 3 plots the *mean* gate value at each recursion step. Under the ponder cost the gates collapse: the multi-task model keeps step 1 open (attention 0.83, FFN 0.66) and closes steps 2–4 to near zero, while reconstruction also concentrates on the first step. The encoder thus discovers that roughly *one* effective recursive application suffices here, with accuracy unchanged (reconstruction stays at 0.10). Two honest caveats. First, we report only the mean gate; to claim the gate is genuinely *input-dependent* (the property that separates it from a per-step learned scalar such as ReZero/LayerScale, or from Mixture-of-Depths [19] routing) one must show across-input variance in the gate and ablate against a non-conditioned per-step scalar, which we have not yet done. Second, a near-zero gate scales the sublayer output but does not skip the FLOPs; the ponder weight therefore trades accuracy for *effective* depth, not measured compute.

### F. The modality gap as a symptom of objective–modality mismatch

Fig. 4 measures alignment quality for the standalone contrastive model: the cosine of matched waveform/constellation pairs is near zero and in fact slightly *negative* (anti-aligned), far from the value of 1 that alignment targets — a stronger statement than "barely above random," since random pairs sit near zero in high dimension. With complementary modalities the shared signal contrastive feeds on is nearly empty, so alignment has almost nothing to converge to. The geometric modality gap is therefore downstream of an objective that wanted the intersection of two modalities whose intersection is thin.

A natural question is whether the multi-task reconstruction loss, by giving the encoder a richer shared representation, lets the contrastive head finally close the gap. Fig. 5 answers it, and the answer is revealing. With an *InfoNCE* head the gap collapses almost completely once reconstruction is added: matched-pair cosine rises from $-0.30$ (standalone) to $+0.96$, and the distance between the two modality centroids falls from 1.61 to 0.12. With a *SigLIP* head it does not: the cosine stays negative ($-0.57 \rightarrow -0.49$) and the centroids stay far apart. Yet the model that closes the gap is the *worse* classifier (recon + InfoNCE trails recon + SigLIP by 10–21 points, Table I). Closing the modality gap and downstream accuracy are thus *decoupled* here. We read this through partial information decomposition, but carefully. The AMC readout pools the *fused* encoder tokens, not the contrastive CLS projections, so the mechanism must act on the shared encoder: the alignment pressure of the InfoNCE head, back-propagated into that encoder, appears to collapse the two modality representations together at the cost of the per-modality *uniqueness* the classifier needs, whereas the looser SigLIP head leaves it intact. We say "appears" deliberately. We have not estimated redundancy/uniqueness/synergy (§VI), and both heads share a single contrastive weight $\lambda_c$=0.1 despite very different loss scales, so part of the effect may be a loss-weight imbalance — InfoNCE's batch softmax exerting stronger gradients than SigLIP's pairwise sigmoid — rather than the loss *family*. Separating these requires a $\lambda_c$ sweep per head at matched alignment strength together with a measured PID, which we leave to the full study.

The bare observation that closing the gap can hurt is not itself new: Liang et al. [36] find the gap–performance relation task-dependent, with driving the gap to zero generally hurting, and Schrodi et al. [37] tie the gap to information imbalance between modalities and to downstream behaviour, while the "contrastive gap" has been recast as a geometric artifact rather than a defect [38]. Our claim is narrower and specific to the RF regime: for complementary modalities the *choice of objective*

— contrastive alignment versus reconstruction — is what governs the representation, and the gap is a *symptom* of an objective–modality mismatch, not a quantity to optimize. The practical rule stands: *align modalities that share information, reconstruct complementary ones.*

### G. Parameter efficiency and a caveat on depth

Sharing one block across $L{=}4$ applications reduces the encoder block parameters by $\approx 4\times$ relative to an untied depth-4 stack at equal width. We state this as an architectural property rather than a validated trade: we did not train the untied baseline, and weight-tied recursion is not capacity-equivalent to distinct layers [12], [13], so an untied comparison at equal width is needed to measure the parameter/accuracy trade. The gate's collapse to one effective step (§IV-E) further cautions against over-reading the recursion contribution on this benchmark: if one step suffices at no accuracy cost, the task does not exercise depth $> 1$, and the parameter saving (which only matters when effective depth $> 1$ is wanted) cannot be demonstrated here. The gate *scales* each sublayer's residual output but does not skip its computation, so it reduces *effective depth*, not FLOPs; turning that into a true compute saving requires hard skipping (threshold the gate and bypass the sublayer) and a latency measurement, which we leave to the full-scale study. We do, however, validate the broader parameter claim on real data in §IV-H, where swapping our recursive encoder for a full ViT-Base in a DenoMAE-2.0 pipeline gives matched accuracy at $\approx 1/11$ of the encoder parameters.

### H. A measured efficiency check on RadioML 2018.01A

To test the parameter claim beyond the synthetic toy, we run the encoder swap on the real RadioML 2018.01A benchmark [1] (24 classes), inside a DenoMAE-2.0-style pipeline: each I/Q frame is rendered to a constellation image, pretrained by masked autoencoding with a local-patch-position auxiliary, and read out by a frozen linear probe — the protocol of [9]. We change *only* the encoder. The ViT-Base encoder of the DenoMAE family has $\approx$ 85.6 M parameters; our recursive encoder has 7.8 M (Table II). Across SNR the two track within 1–3 points and the overall accuracies are 22.4% (ViT-Base) versus 21.2% (recursive), i.e. the recursive encoder retains the ViT-Base accuracy at $\approx 1/11$ of the encoder parameters. We report this as a *preliminary* measurement: pretraining is short and the probe is frozen, so the absolute numbers are modest and a scaled run is in progress; but the architectural conclusion — weight-tied recursion matches a full ViT-Base encoder at a fraction of its parameters — is now measured on real data, not asserted.

## V. Discussion

Partial information decomposition [40] splits the information two modalities carry about a target into redundancy (what either reveals), uniqueness (what only one reveals), and synergy (what only the two together reveal). We use it as a qualitative lens, and state the link carefully: a contrastive objective lower-bounds the mutual information *between the two view embeddings*, not the PID redundancy about $Z$. We also avoid over-reading the bound itself: the representational success of InfoNCE is not cleanly explained by mutual-information maximization [42], so we treat the MI view as motivation, not mechanism. The two coincide only when the information shared between the views is also the information they share about the target; under our construction that shared-between-views information is thin (Fig. 4), so contrastive alignment has little to grip on regardless. The denoising target $Z$, by contrast, lives in the union of the modalities, including the uniqueness and synergy that no agreement-between-views objective is built to capture, so reconstruction is the natural fit. SigLIP does not change this because it alters the loss shape, not the loss family. We summarize a practical rule: *align modalities that share information, reconstruct complementary ones*; the modality gap is a sign of the latter, not a defect to patch.

TABLE II
Encoder swap on RadioML 2018.01A (24 classes, chance 4.2%; frozen linear probe; identical DenoMAE-2.0-style pipeline). The recursive encoder matches ViT-Base at $\approx 1/11$ of the encoder parameters. Preliminary (short pretraining).

| encoder | enc. params | overall acc. | acc. @12 dB |
|---|---|---|---|
| ViT-Base (DenoMAE) | 85.6 M | 22.4% | 41.7% |
| **recursive (ours)** | **7.8 M** | 21.2% | 40.6% |

## VI. Conclusion and Limitations

We combined a recursive (Universal-Transformer/ALBERT) encoder, a CLS-conditioned adaptive-depth gate, and a single-pass multi-task objective for multimodal AMC. The multi-task model gives the best AMC while denoising to half the noise floor; the gate prunes the encoder to about one effective recursive step; and the align-versus-reconstruct analysis explains why reconstruction is the right objective for the complementary modalities AMC uses. We are explicit about what this draft does *not* yet establish. The benchmark is synthetic and small ($d{=}48$); results are single-seed; we report no untied baseline, recursion-depth ($L$) sweep, or FLOP/latency numbers, so the parameter- and compute-efficiency claims are stated, not measured; we report only the mean gate, with no across-input variance or static-gate control, so input-dependence is not yet demonstrated; and the complementarity at the heart of the analysis is argued rather than estimated. Closing these is a concrete program: RadioML 2018.01A (linear-probe and fine-tuned) against DenoMAE and SOTA baselines; an untied-versus-tied $L$-sweep with parameter and FLOP counts; $\geq$ 3 seeds with error bars; a gate input-variance analysis with a static-gate control; a measured partial-information decomposition (redundancy/uniqueness/synergy) of the waveform–constellation pair [40]; and an on-device latency study with hard gate skipping.

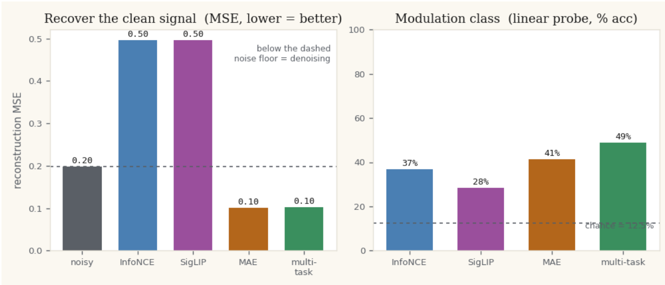


Fig. 1. Left: recovering the clean signal. Both contrastive losses stay at 0.50, above the noise floor (dashed); reconstruction and multi-task denoise to 0.10. Right: linear-probe modulation class, where the multi-task model leads.

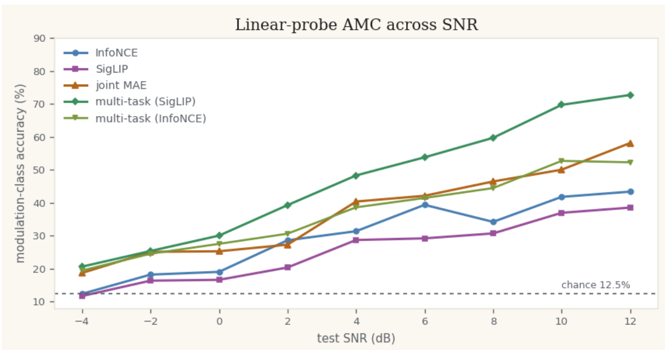


Fig. 2. Linear-probe AMC accuracy versus SNR. The SigLIP-head multi-task model leads across the range, reconstruction and the InfoNCE-head multi-task model follow, and the standalone contrastive encoders trail.


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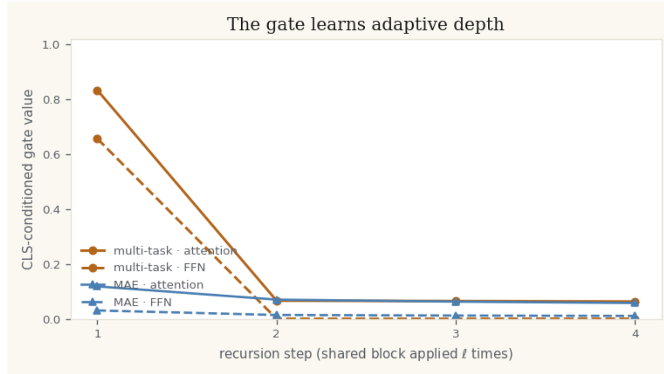


Fig. 3. Mean CLS-conditioned gate value at each recursion step. Under the ponder cost the gates collapse: the multi-task model keeps step 1 open and closes steps 2–4, so the encoder learns it needs about one effective recursive application.

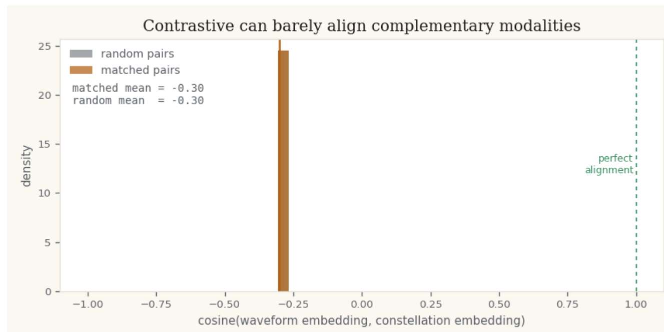


Fig. 4. Contrastive alignment quality. Matched modality pairs reach a cosine far from 1, barely above random pairs: complementary modalities give alignment little to grip.


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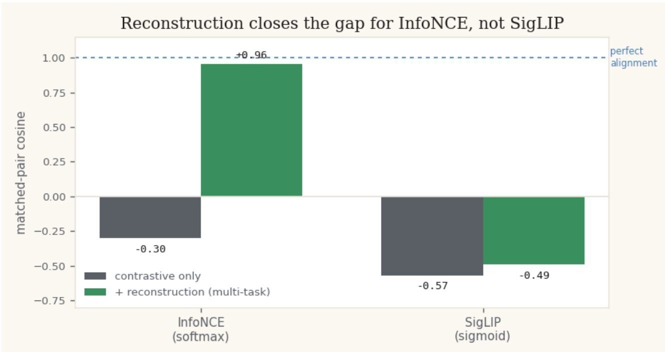


Fig. 5. Does reconstruction close the modality gap? Matched-pair cosine for each contrastive head, standalone versus as a multi-task auxiliary to reconstruction. The InfoNCE head closes the gap (to $+0.96$); the SigLIP head does not — yet the SigLIP head is the better classifier (Table I), so closing the gap does not help.

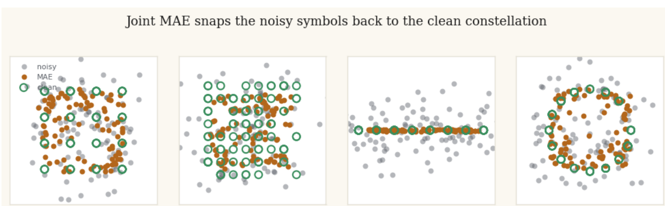


Fig. 6. The recursive MAE snaps noisy symbols (gray) back toward the clean constellation (green); reconstruction in orange.


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## Appendix

We instantiate the two modalities, in the spirit of the DenoMAE family [8], [9], as explicit lossy projections of the clean signal. Let $Z$ be a length-$M$ sequence of symbols drawn from one of $K$ modulation schemes. The receiver observes

$$X_{\text{wav}} = Z + \eta, \qquad X_{\text{con}} = \text{Hist}_{G\times G}(Z + \eta), \tag{4}$$

where $\eta$ is additive white Gaussian noise at a given SNR. The waveform $X_{\text{wav}}$ keeps symbol order but is noisy; the constellation $X_{\text{con}}$, a 2D I/Q histogram, keeps clean cluster geometry but discards order. Recovering $Z$ therefore needs order from the waveform and geometry from the constellation, so the two views are complementary rather than redundant; this is the structural fact the align-versus-reconstruct analysis rests on.

The benchmark draws $K{=}8$ schemes (BPSK, QPSK, 8PSK, 16QAM, 64QAM, 4PAM, 8PAM, 16PSK) with $M{=}96$ symbols per example and a $G{=}24$ constellation histogram. Representations are pretrained on signals at SNR sampled uniformly from $[-2, 10]$ dB and evaluated from $-4$ to $12$ dB, matching the DenoMAE-style across-SNR AMC protocol of Sec. IV-A.

Our contrastive variants are the standard CLIP and SigLIP losses, used unmodified and restated here for completeness. Given L2-normalized per-modality `[CLS]` embeddings $\boldsymbol{z}^{\text{wav}}, \boldsymbol{z}^{\text{con}}$, the softmax InfoNCE loss of CLIP [34] is

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{N}\sum_i \log \frac{\exp(\boldsymbol{z}_i^{\text{wav}} \cdot \boldsymbol{z}_i^{\text{con}}/\tau)}{\sum_j \exp(\boldsymbol{z}_i^{\text{wav}} \cdot \boldsymbol{z}_j^{\text{con}}/\tau)}, \tag{5}$$

with temperature $\tau$. The SigLIP [35] pairwise-sigmoid alternative is

$$\mathcal{L}_{\text{SigLIP}} = \sum_{i,j} \text{softplus}\big(-y_{ij}\,(t\,\boldsymbol{z}_i^{\text{wav}} \cdot \boldsymbol{z}_j^{\text{con}} + b)\big), \tag{6}$$

with $y_{ii} = 1$, $y_{i\neq j} = -1$ and learned scale and bias $t, b$. In the multi-task objective $\mathcal{L}_{\text{SigLIP}}$ is the contrastive term scaled by $\lambda_c$.